Comparitive Variant Analysis Workflow for Pearl Millet Rancidity Genotypes

1 Variety

GHB - 538 (high rancid)



GHB - 732 (low rancid)



2 Variant calling

GHB - 538 (high rancid)

Initial total SNPs and INDELs: **17,842,802**

77.3%

Final total SNPs and INDELs: **13,799,108**

GHB - 732 (low rancid)

Initial total SNPs and INDELs: **18,686,050**

80.88%

Final total SNPs and INDELs: **15,112,744**

3 Variant breakdown

GHB - 538 (high rancid)

89.54% SNPs

10.46% INDELs

Final SNPs: **12,355,115**

Final INDELs: **1,443,993**

GHB - 732 (low rancid)

90.14% SNPs

9.86% INDELs

Final SNPs: **13,622,513**

Final INDELs: **1,490,251**

4 Genomic effect of Variants

GHB - 538 (high rancid)

35,772 classified gene id entry

- Gene having only SNP = **284**
- Gene having only INDEL = **21**
- Gene having SNP + INDEL = **35,466**

GHB - 732 (low rancid)

35,745 classified gene id entry

- Gene having only SNP = **280**
- Gene having only INDEL = **34**
- Gene having SNP + INDEL = **35,431**

5 Variant effect categories

Effect category	GHB538 (High rancid)	GHB732 (Low rancid)
Loss of function	**13,336**	**10,181**
Protein-altering	**15,792**	**16,584**
Regulatory	**4,300**	**8,965**
Low impact coding	**2,343**	**14**
Compound gene ID	**1**	**1**

7 Rancidity-associated genes

Summary	GHB538 (High rancid)		GHB 732 (low rancid)	
	Category	Number of genes	Category	Number of genes
Total genes	1108		585	
	Protein-altering	497	Protein-altering	293
	LoF	419	LoF	424
	Regulatory	98	Regulatory	143
	Low impact coding	94	Low impact coding	0

6 Functional Annotation summary (%)

Annotation category	GHB538 (High rancid)	GHB732 (Low rancid)
LoF	**80.53%**	**74.13%**
Protein-altering	**87.31%**	**85.00%**
Regulatory	**84.67%**	**89.55%**
LOw impact coding	**91.34%**	**100.00%**
Compound gene ID	**100%**	**100%**
Overall annotation	**90.60%**	**83.05%**